

AJAY VIKRAM PERIASAMI

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Education

Duke University

M.S. in Computer Science (AI/ML specialization)
GPA: 4.0/4.0

North Carolina, USA
Aug 2025 – May 2027

National Institute of Technology Karnataka

B.Tech. in Computer Science and Engineering
GPA: 9.46/10 · Department Rank: 4/130

Surathkal, India
Dec 2020 – Apr 2024

Research Interests

Embodied intelligence; studying how robots can learn the most useful internal abstractions of the physical world for planning, reasoning, action and failure recovery, with a focus on multimodal world models that capture dynamics, affordances, and action consequences across vision, touch, audio, language, and proprioception, supporting generalization to unfamiliar tasks.

Research Experience

The Robot Dexterity Lab, Duke University

Graduate Researcher · Advisor: Xianyi Cheng

May 2026 – Present

- Developed **FRAMES**, a supervisory framework for Unitree G1 humanoid loco-manipulation combining LLM planning and recovery, skill-level semantic memory, and SAM3/RGB-D object grounding over a fixed whole-body controller.
- Built a VLM monitor using temporal multi-view images, robot state, and contact evidence to interrupt failed skills and guide recovery. Independent evaluation on 100 MuJoCo trials across five failure categories achieved 94% classification accuracy, detecting 48/50 failures and correctly accepting 46/50 successful executions. Further extensions are ongoing.

DukeNLP, Duke University

Graduate Researcher · Advisor: Bhuwan Dhingra

Sep 2025 – Present

- Built **Vision2Code**, a reference-code-free image-to-code benchmark spanning 539 test examples, 15 datasets, and six domains. Dataset-specific VLM rubrics achieved up to 0.861 Spearman correlation with four human annotator ratings.
- Benchmarked nine VLMs and developed evaluator-guided self-training, improving Qwen3.5-9B from 1.60 to 1.86 and Phi-4-MM from 0.58 to 0.74 on a 0–5 reconstruction scale, with gains confirmed by two additional VLM judges.
- Developing a multimodal math tutor with progressive visual hints. Exploring interactive animations and learning-gap inference from responses, and designing a benchmark with simulated learners, evaluation rubrics, and human validation.

Duke Institute for Brain Sciences, Duke University

Graduate Research Assistant · Advisor: Andrew M. Michael

Sep 2025 – Present

- Developed a 3D U-Net for fMRI motion correction without motion regressors at inference, improving mean voxelwise correlation with fully preprocessed references from 0.6413 to 0.9368 on held-out test subjects with synthetic motion.
- Developing latent-space fMRI models for downstream phenotype evaluation and assessing reliability across runs.

Rhodes Information Initiative, Duke University

Data+ Researcher · Advisor: Andrew M. Michael

May – Jul 2026

- Investigated structure–function coupling in 896 HCP participants using white-matter fractional anisotropy and resting-state fMRI. Compared structural, functional, and combined regression models using partial least squares for dimensionality reduction, finding that combined features improved age and fluid-intelligence prediction over fMRI alone.

NeuRonICS Lab, Indian Institute of Science

Machine Learning Research Intern · Advisor: Chetan Singh Thakur

May 2024 – Jul 2025

- Built asynchronous trackers for astronomical objects and drones by clustering neuromorphic events. Prototyped in Python and optimized C code with compact data structures and multithreading for real-time ARM Cortex-A72/M7 inference.
- Compressed CNNs and state-space models (SSMs) using quantization, ternarization, pruning, and mixed precision. Ternary weights enabled $16\times$ raw weight compression with under 1% accuracy loss on radar activity recognition.
- Benchmarked SSMs like S4, Mamba, and LMU for gesture, speech emotion, and human activity recognition, and studied HiPPO representations. Achieved $R^2 = 0.75$ using S4 to decode 2D hand trajectories from primate neural recordings.
- Designed low-latency LSTM models for primate neural decoding, ranking 5th in the IEEE BioCAS 2024 Grand Challenge.

National Institute of Technology Karnataka

Undergraduate Researcher · Advisor: Jeny Rajan

Oct 2023 – Mar 2024

- Developed **ViTFuser** for end-to-end autonomous driving with multi-resolution ViT fusion of RGB and LiDAR features and a Feature Pyramid Network for improved multi-scale object detection. Achieved a CARLA Longest6 driving score of 55.15 versus 43.48 for the TransFuser baseline, with 67% fewer parameters and one-third the GPU memory use.
- Performed a literature survey on breast cancer detection, extracted regions of interest (ROIs) from mammograms, and experimented with ViT and Swin Transformer models on the resulting patches, achieving 70% classification accuracy.

Publications

Conference Papers

- [1] **Asynchronous High-Speed Tracking of Astronomical Objects Using Neuromorphic Camera for Edge Computing.**
Satyapreet Singh Yadav, Ajay Vikram P, Adithya M D, Chandra Sekhar Seelamantula, Chetan Singh Thakur.
IEEE ICASSP 2026 — Oral [paper]
- [2] **ViTFuser: Advancements in Global Context Understanding for Autonomous Vehicles.**
Atanu Chatterjee, Ajay Vikram P, Aniketh Narayan Bellala, T. Naga Tarun, Basavaraj Talawar, M. Vani, Jeny Rajan.
MIND 2024 [paper]

Workshop Papers

- [3] **FRAMES: Failure Recovery And Monitoring of Embodied Skills for Humanoid Loco-Manipulation.**
Ajay Vikram P, Xinyuan Luo, Haoyu Li, Xianyi Cheng.
Full-Shift Robot Co-Workers Workshop, IROS 2026 [paper]
- [4] **Reinforcement Learning for the Game of Balatro: A Fast Simulator and Two-Policy Training Approach.**
Ajay Vikram P, Anjaneya Gupta, Ethan Cheng, Ivan Rosales.
Experimental AI in Games Workshop, AAAI AIIDE 2026

Under Review

- [5] **Vision2Code: A Multi-Domain Benchmark for Evaluating Image-to-Code Generation.**
Ajay Vikram P, Junlin Wang, Bhuwan Dhingra.
ICLR 2027 — Under Review [paper]
- [6] **A Novel Deep Learning Approach to fMRI Head Motion Correction without Motion Regressors.**
Ajay Vikram P, Bharat Kashyap Karri, Andrew M. Michael.
IEEE BIBM 2026 — Under Review
- [7] **HOMI: Ultra-Fast EdgeAI platform for Event Cameras.**
Shankaranarayanan H, Satyapreet Singh Yadav, Adithya Krishna, Ajay Vikram P, Mahesh Mehendale, Chetan Singh Thakur.
IEEE Internet of Things Journal — Under Review [paper]

Posters & Abstracts

FRAMES: Failure Recovery And Monitoring of Embodied Skills for Humanoid Loco-Manipulation.

IROS 2026 Workshop on Full-Shift Robot Co-Workers — Poster presentation

fMRI Measures of the Human Brain and Associations to Behavior.

Duke Neurology Research Symposium, 2026 — Poster presentation

Structure-Function Coupling in the Human Brain and Links to Behavior.

Duke Neurology Research Symposium, 2026 — Poster presentation

Comparison of DMN and other Brain Networks during resting fMRI.

IEEE EMBC 2026 — Accepted abstract

Neuromorphic Camera for Event-Based Motion Detection and Tracking.

Indian Institute of Science Open Day, 2025 — Poster presentation and live demonstration

Selected Research Projects

Multi-Task LLM Post-Training

Feb – Apr 2026

- Developed a four-stage LoRA pipeline for Llama-3.1-8B combining task-aligned SFT with GRPO-style RL using verifiable rewards and analyzed how data mixtures, response formats, and reward sparsity affect instruction following, math, and code. Achieved a mean score of 72.3% across IFEval, GSM8K, and HumanEval, ranking 2nd among 22 course teams.

SkillBlenderLC

Sep – Dec 2025

- Developed an LLM-guided extension of SkillBlender for humanoid skill acquisition, linking task decomposition and missing-skill detection to executable reward synthesis and PPO training in Isaac Gym. Benchmarked six LLMs across eight tasks (GPT-4o: 75% strict skill-selection accuracy) and identified reward-scaling and balance limitations in trained policies.

Event-Based Drone Tracking

Sep – Dec 2024

- Developed a non-neural drone tracker using event clustering and Hungarian matching; maintained the selected track in 198/200 windows over a 10-second segment and optimized the event processor in C for deployment on Raspberry Pi 4.

Ternarizing CNNs for Human Activity Recognition

Jun – Aug 2024

- Trained a 3,029-parameter CNN on radar spectrograms with ternarized weights and 8-bit quantized activations using quantization-aware training. Combined pruning to approximately 45% sparsity, batch-normalization folding, and dyadic requantization for integer-only inference, achieving 98.91% mean accuracy within 0.51 percentage points of FP32.

Teaching

Probabilistic Machine Learning, Duke University

Jan – Apr 2026

Graduate Teaching Assistant

- Led lab sessions, held office hours, provided project guidance, and graded assignments.

Achievements

17th place, IEEE ICASSP EEG Auditory Attention Decoding Challenge

2026

Contribution: efficient state-space models for auditory attention decoding.

Dean's Scholarship, Boston University

2025

Offered \$20,000 for the M.S. in Artificial Intelligence.

5th place, IEEE BioCAS Grand Challenge on Neural Decoding

2024

Contribution: efficient LSTM models for primate neural decoding.

Annual Proficiency Prize for Academic Excellence, Bishop Cotton Boys' School

2011-2018

Seven consecutive awards for outstanding academic performance.

Industry Experience

Lamarr

Mar – Apr 2024

Machine Learning Intern · Remote

- Integrated Table Transformer and TrOCR for scanned documents and handwritten text; fine-tuned Qwen1.5-14B on custom datasets using Llama Factory to tailor the model responses to domain-specific legal and business requirements.
- Developed Draft Bot to retrieve document templates and fill customizable placeholders for automated document preparation; built a SalesGPT agent to schedule automated service-promotion emails and streamline client communication.

Qualcomm

May – Jul 2023

Software Intern · Hyderabad, India

- Implemented a Python-based diff engine to compare memory usage across builds, surfacing regressions and optimizations.
- Built and optimized a memory-analysis tool, cutting processing time by 85% and memory usage by 75%; streamlined the end-to-end analysis workflow to require a single changelist ID as user input, with automated email notifications for users.
- Developed a React interface featuring interactive memory graphs, filtering, sorting, and tree-based memory organization, enabling users to view analysis results, navigate memory hierarchies, and interpret memory usage through visualizations.
- Automated CI/CD with Jenkins, Docker, and Kubernetes, including load balancing and user authentication; the tool's gains in processing time and memory efficiency and successful adoption across the team led to a full-time employment offer.

Leadership & Service

Member, Computer Vision Team, Duke Robotics Club

Aug – Dec 2025

Worked on object detection and performance profiling on NVIDIA Jetson Nano.

Bangalore Neuromorphic Engineering Workshop, IISc

Jan 2025

Organizer & Participant: assisted with organization and participated as a student researcher.

Placement Coordinator, Career Development Center, NITK

Aug 2023 – Apr 2024

Coordinated campus recruitment for full-time roles and internships across companies; organized programs and events to develop students' technical skills based on industry feedback.

Volunteer Mentor, Leadership Development, Team Everest NGO

Feb – Mar 2024

Mentored an underprivileged student 1:1 in problem solving, decision making, active listening, character, communication, empathy, honesty, and conflict resolution.

Relevant Coursework

Duke University: Theory & Algorithms of Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Natural Language Processing, Advanced Machine Learning, High Dimensional Statistics, Building Intelligent Agents.

NIT Karnataka: Linear Algebra & Matrices, Discrete Math, Machine Learning, Digital Image Processing, Big Data Analytics.

Technical Skills

Programming: Python, C/C++, MATLAB, Rust, JavaScript, SQL.

Machine learning: PyTorch, TensorFlow, scikit-learn, Brevitas, Llama Factory.

RL & robotics: Gymnasium, Ray RLlib, MuJoCo, ROS 2, CARLA.

Scientific computing: NumPy, pandas, OpenCV.

Systems & tooling: Linux, Git, Docker, Kubernetes, Jenkins, MLflow.